Eider Aldana, Ph.D. Candidate

eider.aldana2201@alumnos.ubiobio.cl
ResearchGate
Orcid
Linkedin
Research area: Numerical Analysis of Partial Differential Equations



Personal Information

R.U.N.	27.760.351-9
Address	Pedro del Río Zañartu 1045, Concepción, Chile
Email	eider.aldana2201@alumnos.ubiobio.cl
Phone	+56 961852531

Education

2022 – 2026 **Ph.D. in Applied Mathematics, Universidad del Bío-Bío, Chile**.
Research area: Numerical Analysis of Partial Differential Equations.
Thesis topic: Numerical methods and finite element discretizations for coupled PDE systems arising in fluid dynamics.

2017 – 2021 **M.Sc. in Mathematics, Universidad del Atlántico, Colombia**.
Thesis title: *(The spectrum of the power graph of a cyclic $p-$group and some characteristics of an orthogonal graph in an indefinite metric space).*

2013 – 2017 **B.Sc. in Mathematics, Universidad de Sucre, Colombia**.

Skills

Languages	Spanish (native); English (B2, Upper-Intermediate, CEFR; academic and professional working proficiency).
Programming	Python, MATLAB, FreeFem++, MPI (parallel computing), LaTeX.
Scientific Computing	Numerical methods for partial differential equations, finite element methods, numerical simulation of fluid dynamics problems.
Software	Linux, macOS, Git, scientific computing environments.
Techniques	Finite element methods, variational formulations of PDEs, numerical discretization of coupled systems (Navier–Stokes and convection–diffusion), stability and convergence analysis of numerical schemes.
Misc.	Academic research in numerical analysis of PDEs, university teaching, preparation of mathematical teaching materials, LaTeX typesetting and scientific writing.

Teaching Experience

During this period, I have had the opportunity to teach both theoretical and practical classes in several Engineering programs at the Universidad del Bío-Bío. This experience has allowed me to develop innovative teaching strategies in which I incorporate technological resources to reinforce theoretical concepts and present abstract mathematical ideas in an engaging and accessible way. The courses I have taught are:

2018 **Universidad de Sucre**. Sincelejo, Sucre, Colombia. Academic Monitor in Mathematics II, one semester.

Teaching Experience (continued)

2019 — **Universidad de Sucre**. Sincelejo, Sucre, Colombia. Academic Monitor in Calculus II, one semester.

2019–2021 — **Instituto Técnico Cultural Diocesano**. Magangué, Bolívar, Colombia. Full-time Lecturer.

2021 — **Universidad de Sucre**. Sincelejo, Sucre, Colombia. Lecturer for *Set Theory* and *Mathematics I*, one semester.

2022 — **Universidad de Cartagena**. Magangué, Bolívar, Colombia. Lecturer, four months.

2022–2023 — **Corporación Universitaria del Caribe (CECAR)**. Sincelejo, Sucre, Colombia. Lecturer for *Descriptive Statistics* and *Inferential Statistics*, one year.

2025–2 — **Universidad del Bío-Bío / Teaching Assistant.** "Calculus I" (Course Code: 220157), Bachelor of Engineering in Computer Science and Information Technology, 6 hours per week.

2026–1 — **Universidad del Bío-Bío / Teaching Assistant.** "Algebra I" (Course Code: 220155), Bachelor of Engineering in Computer Science and Information Technology, 8 hours per week.

Universidad del Bío-Bío / Teaching Assistant. "Differential Calculus" (Course Code: 220295), Electrical Engineering, 6 hours per week.

Universidad del Bío-Bío / Teaching Assistant. "Algebra and Trigonometry" (Course Code: 220296), Electronic Engineering, 6 hours per week.

2026–2 — **Universidad del Bío-Bío / Teaching Assistant.** "Differential Equations" (Course Code: 220135), Civil Engineering, 6 hours per week.

Universidad del Bío-Bío / Teaching Assistant. "Calculus I" (Course Code: 220120), Business Engineering, 6 hours per week.

Universidad del Bío-Bío / Teaching Assistant. "Differential and Integral Calculus" (Course Code: 220322), 8 hours per week.

Interships and Fellowships

2023 – 2026 — **ANID Scholarship – Doctorado Becas Chile 2023** (Folio 21230951). Eider Aldana is partially supported by ANID-Chile through the Scholarship Program, Doctorado Becas Chile 2023.

Sep – Dec 2024 — **Research Visit, Department of Mathematics, University of North Texas, USA**. Research stay conducted as part of the Ph.D. program in Applied Mathematics at Universidad del Bío-Bío, working on the thesis topic *Numerical Analysis for a Reverse Osmosis Model*, under the supervision of **Prof. Giordano Tierra Chica**.

2026 — **Travel Grant for Attendance at Scientific Events 2025**. Vice-Rectorate for Research and Graduate Studies, Universidad del Bío-Bío, Chile.

Attendance at Scientific Events

Conference Talks

2026 — **XXIX Congreso de Ecuaciones Diferenciales y Aplicaciones (CEDYA) – XIX Congreso de Matemática Aplicada (CMA)**. Spain. Presentation titled: *Numerical Analysis of a three-field formulation for a Reverse Osmosis Model*.

XXXV Jornada de Matemática de la Zona Sur (JMZS). Universidad del Bío-Bío, Concepción, Chile. Presentation titled: *A three-field formulation for an unsteady Reverse Osmosis model*.

2025 — **Escuela de Primavera en Matemática Aplicada (E'(UBB))**. Universidad del Bío-Bío, Concepción, Chile. Presentation titled: *Numerical Analysis for a Reverse Osmosis Model*.

Attendance at Scientific Events (continued)

- **XCIII Encuentro Anual de la Sociedad de Matemática de Chile (SOMACHI)**. Universidad de La Serena, Vicuña, Chile, November 25-28, 2025. Presentation titled: *Numerical Analysis of a three-field formulation for a Reverse Osmosis Model*.
- **SIAM Conference Texas–Louisiana Section, 7th Annual Meeting**. Baylor University, Waco, Texas, USA. Presentation titled: *Analysis of a Conforming Finite Element Method for a Reverse Osmosis Model*.
- **Tenth Meeting on Numerical Analysis of Partial Differential Equations (Talca Numérica I)**. Universidad Católica del Maule, Talca, Chile. Talk titled: *Analysis of a Conforming Finite Element Method for a Reverse Osmosis Model*. Book of abstracts available at: `https://ci2ma.udec.cl/TN/book-of-abstracts.pdf`.

2022
- **Primer Encuentro Nacional de Estudiantes de Matemática (ENEMAT)**. Pontificia Universidad Católica de Chile, November 2–4, 2022.

2024
- **Seventh Chilean Workshop on Numerical Analysis of Partial Differential Equations** (WONAPDE 2024).

2023
- **Primer Workshop de Matemática Multidisciplinaria**. Co-organized by Universidad del Bío-Bío and Universidad de Concepción. Participation also in the First and Second Workshop on Multidisciplinary Mathematics, Universidad del Bío-Bío.
- **XXXV Jornada de Matemática de la Zona Sur (JMZS)**. Concepción, Chile.

Scientific Event Organization

2025–Present
- **Seminar on Numerical Analysis of the Bío-Bío (SABIO)**. Universidad del Bío-Bío, Concepción, Chile. Co-organizer since July 10, 2025. The seminar aims to foster the exchange of ideas and recent research results in numerical analysis and its applications, bringing together researchers, students, and professionals through regular scientific meetings. News page: `https://noticias.ubiobio.cl/`.

Research Publications

1. **E. Aldana** and R. Oyarzúa, *A lagrange multiplier approach for fluid-flow problem with inhomogeneous mixed boundary conditions*, Preprint, CI^{2}MA, Universidad de Concepción, 2026. URL: `https://www.ci2ma.udec.cl/publicaciones/prepublicaciones/prepublicacion.php?id=631`
2. **E. Aldana** and R. Oyarzúa, "A lagrange multiplier-based finite element method for a reverse osmosis model," Centro de Investigación en Ingeniería Matemática (CI²MA), Universidad de Concepción, Tech. Rep. Pre-Publicación 2026-33, 2026. URL: `https://www.ci2ma.udec.cl/publicaciones/prepublicaciones/prepublicacion.php?id=645`
3. **E. Aldana**, R. Oyarzúa, and J. Vellojín, "Numerical analysis of a three-field formulation for a reverse osmosis model," *IMA Journal of Numerical Analysis*, 2026. DOI: `10.1093/imanum/drag065`

References

Ricardo Oyarzúa, Departamento de Matemática, Universidad del Bío-Bío, Concepción, Chile, email: royarzua@ubiobio.cl

Giordano Tierra Chica, Department of Mathematics, University of North Texas, Denton, Texas, USA, email: giordano.tierrachica@unt.edu